

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Symmetric Algebraic Integral

Evaluate the definite integral:

$$\int_{1/2}^2 \frac{x^4 - 1}{x^2 \sqrt{x^4 + 1}} dx$$

Solution: Let I be the integral. We can rewrite the integrand by splitting the fraction and manipulating the terms:

$$I = \int_{1/2}^2 \frac{x^4 - 1}{x^2 \sqrt{x^4 + 1}} dx = \int_{1/2}^2 \frac{x^2 - x^{-2}}{\sqrt{x^4 + 1}} dx$$

To reveal the hidden symmetry, let's divide the numerator and the terms inside the square root by x :

$$I = \int_{1/2}^2 \frac{x - x^{-3}}{\sqrt{x^2 + x^{-2}}} dx$$

Let's apply the substitution $x = \frac{1}{t}$. This gives $dx = -\frac{1}{t^2} dt$. The bounds change from $[1/2, 2]$ to $[2, 1/2]$. Substitute these into the integral:

$$I = \int_2^{1/2} \frac{(1/t)^4 - 1}{(1/t)^2 \sqrt{(1/t)^4 + 1}} \left(-\frac{1}{t^2}\right) dt$$

Simplify the expression inside the integral:

$$\begin{aligned} I &= \int_2^{1/2} \frac{(1 - t^4)/t^4}{1/t^2 \sqrt{(1 + t^4)/t^4}} \left(-\frac{1}{t^2}\right) dt \\ I &= \int_2^{1/2} \frac{1 - t^4}{t^2 \sqrt{t^4 + 1}} \left(-\frac{1}{t^2}\right) dt = \int_{1/2}^2 \frac{1 - t^4}{t^4 \cdot \frac{1}{t^2} \sqrt{t^4 + 1}} dt \\ I &= \int_{1/2}^2 \frac{1 - t^4}{t^2 \sqrt{t^4 + 1}} dt = - \int_{1/2}^2 \frac{t^4 - 1}{t^2 \sqrt{t^4 + 1}} dt \end{aligned}$$

Notice that this resulting expression is exactly the negative of our original integral I . Therefore:

$$I = -I \implies 2I = 0 \implies I = 0$$

The value of the definite integral is exactly 0.

1.2 Hard Integral

Hyperbolic Zeta Integral

Evaluate the definite integral:

$$\int_0^\infty \frac{\tanh^2(x)}{x^2} dx$$

Solution: Let $I = \int_0^\infty \frac{\tanh^2(x)}{x^2} dx$. We utilize the infinite series expansion for $\frac{\tanh(x)}{x}$ obtained via the poles of the hyperbolic tangent function:

$$\frac{\tanh(x)}{x} = \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2 + (2x/\pi)^2}$$

Squaring this expression gives:

$$\frac{\tanh^2(x)}{x^2} = \frac{64}{\pi^4} \left(\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2 + (2x/\pi)^2} \right)^2$$

To integrate this, let us apply the substitution $y = \frac{2x}{\pi} \implies dx = \frac{\pi}{2} dy$. The integral transforms into:

$$I = \frac{32}{\pi^3} \int_0^\infty \left(\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2 + y^2} \right)^2 dy$$

Expanding the squared sum into a double series:

$$I = \frac{32}{\pi^3} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \int_0^\infty \frac{1}{((2n-1)^2 + y^2)((2m-1)^2 + y^2)} dy$$

We evaluate the standard rational integral $\int_0^\infty \frac{1}{(a^2+y^2)(b^2+y^2)} dy = \frac{\pi}{2ab(a+b)}$ where $a = 2n-1$ and $b = 2m-1$. Substituting this evaluation into our series:

$$I = \frac{32}{\pi^3} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{\pi}{2(2n-1)(2m-1)(2n+2m-2)}$$

$$I = \frac{16}{\pi^2} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{1}{(2n-1)(2m-1)(2n+2m-2)}$$

Notice that $(2n-1)$ and $(2m-1)$ span all positive odd integers. We can rewrite the sum as:

$$I = \frac{16}{\pi^2} \sum_{j,k \text{ odd}} \frac{1}{j \cdot k \cdot (j+k)}$$

Using the algebraic identity $\frac{1}{j \cdot k} = \frac{1}{j+k} \left(\frac{1}{j} + \frac{1}{k} \right)$, the sum simplifies:

$$\sum_{j,k \text{ odd}} \frac{1}{j \cdot k \cdot (j+k)} = \sum_{j,k \text{ odd}} \frac{1}{(j+k)^2} \left(\frac{1}{j} + \frac{1}{k} \right) = 2 \sum_{j,k \text{ odd}} \frac{1}{j(j+k)^2}$$

It is a known beautiful identity of multiple zeta values that this specific double sum evaluates to $\frac{7}{8}\zeta(3)$. Thus, substituting this value back:

$$I = \frac{16}{\pi^2} \left(\frac{7}{8}\zeta(3) \right) = \frac{14}{\pi^2}\zeta(3)$$

The exact value of the integral is $\frac{14}{\pi^2}\zeta(3)$.

Differentials

2.1 Medium Differential

Second-Order Linear ODE

Consider the following differential equation:

$$y''(x) - 9y'(x) + 18y(x) = 9$$

Given that $y(0) = 1$ and $y'(0) = 9$, find the value of $y(\ln(\sqrt[3]{2}))$.

Solution: First, we find the general solution to the homogeneous equation $y'' - 9y' + 18y = 0$. The characteristic equation is:

$$r^2 - 9r + 18 = 0 \implies (r - 3)(r - 6) = 0$$

The roots are $r = 3$ and $r = 6$. The homogeneous solution is:

$$y_h(x) = C_1 e^{3x} + C_2 e^{6x}$$

For the particular solution $y_p(x)$, since the right-hand side is a constant (9), we guess $y_p(x) = A$. Substituting into the ODE gives $18A = 9 \implies A = 1/2$. The general solution is therefore:

$$y(x) = C_1 e^{3x} + C_2 e^{6x} + \frac{1}{2}$$

Next, we apply the initial conditions to find C_1 and C_2 : 1) $y(0) = 1 \implies C_1 + C_2 + \frac{1}{2} = 1 \implies C_1 + C_2 = \frac{1}{2}$ 2) $y'(x) = 3C_1 e^{3x} + 6C_2 e^{6x} \implies y'(0) = 3C_1 + 6C_2 = 9$ Dividing the second equation by 3 gives:

$$C_1 + 2C_2 = 3$$

Subtract the first equation from this result:

$$(C_1 + 2C_2) - (C_1 + C_2) = 3 - 0.5 \implies C_2 = 2.5 = \frac{5}{2}$$

Substituting C_2 back into the first equation:

$$C_1 + 2.5 = 0.5 \implies C_1 = -2$$

The exact solution is:

$$y(x) = -2e^{3x} + \frac{5}{2}e^{6x} + \frac{1}{2}$$

Now, evaluate the function at $x = \ln(\sqrt[3]{2}) = \frac{1}{3} \ln(2)$:

$$e^{3x} = e^{\ln(2)} = 2 \quad \text{and} \quad e^{6x} = (e^{3x})^2 = 4$$

$$y\left(\ln(\sqrt[3]{2})\right) = -2(2) + \frac{5}{2}(4) + \frac{1}{2} = -4 + 10 + 0.5 = 6.5$$

The value is $\frac{13}{2}$ or 6.5.

2.2 Hard Differential

Lagrange Differential Equation

Solve the following first-order Lagrange differential equation:

$$y = e^{y'} - x(y')^2$$

Assuming that $x = x(p)$, $y = y(p)$ are solutions where $p = y'$ and using given conditions $x(-2) = e^{-2}$, and approximations $\text{Ei}(-2) = 0$, $\text{Ei}(1) = 2$, compute the exact value of $x(1) + y(1)$.

Solution: Let $p = y' = \frac{dy}{dx}$. The equation can be written as:

$$y = e^p - xp^2$$

Differentiate both sides with respect to x :

$$p = \frac{dy}{dx} = e^p \frac{dp}{dx} - p^2 - 2xp \frac{dp}{dx}$$

Rearrange to group $\frac{dp}{dx}$ terms:

$$p(p+1) = (e^p - 2xp) \frac{dp}{dx} \implies \frac{dx}{dp} = \frac{e^p - 2xp}{p(p+1)} = \frac{e^p}{p(p+1)} - \frac{2}{p+1}x$$

This is a first-order linear ordinary differential equation for $x(p)$:

$$\frac{dx}{dp} + \frac{2}{p+1}x = \frac{e^p}{p(p+1)}$$

The integrating factor is $\mu(p) = \exp\left(\int \frac{2}{p+1} dp\right) = (p+1)^2$. Multiplying through yields:

$$\frac{d}{dp} [x(p+1)^2] = \frac{e^p}{p(p+1)}(p+1)^2 = \frac{e^p(p+1)}{p} = e^p + \frac{e^p}{p}$$

Integrate both sides with respect to p :

$$x(p+1)^2 = \int \left(e^p + \frac{e^p}{p} \right) dp = e^p + \text{Ei}(p) + C$$

Using the initial condition $x(-2) = e^{-2}$ and $\text{Ei}(-2) = 0$:

$$e^{-2}(-2+1)^2 = e^{-2} + 0 + C \implies e^{-2} = e^{-2} + C \implies C = 0$$

Thus, $x(p)$ is exactly:

$$x(p) = \frac{e^p + \text{Ei}(p)}{(p+1)^2}$$

We want to evaluate $x(1) + y(1)$. First, evaluate $x(1)$ using $\text{Ei}(1) = 2$:

$$x(1) = \frac{e^1 + 2}{(1+1)^2} = \frac{e+2}{4}$$

From the original equation parameterized by p , we have $y(p) = e^p - x(p)p^2$. At $p = 1$:

$$y(1) = e^1 - x(1)(1)^2 = e - \frac{e+2}{4} = \frac{4e - e - 2}{4} = \frac{3e - 2}{4}$$

Finally, compute the requested sum:

$$x(1) + y(1) = \frac{e+2}{4} + \frac{3e-2}{4} = \frac{4e}{4} = e$$

The exact value of $x(1) + y(1)$ is e .

Derivatives

3.1 Medium Derivative

Towering Trigonometric Derivative

Let $f(x) = (\sin(x))^{\cos(x)} \tan(x)$. Find $f'(\pi)$.

Solution: Let $g(x) = (\cos(x))^{\tan(x)}$. Then our function is $f(x) = (\sin(x))^{g(x)}$. We rewrite $f(x)$ using the natural exponential and logarithm:

$$f(x) = \exp(g(x) \ln(\sin x))$$

Applying the chain rule and product rule to differentiate:

$$f'(x) = f(x) \frac{d}{dx} (g(x) \ln(\sin x)) = (\sin x)^{g(x)} \left[g'(x) \ln(\sin x) + g(x) \frac{\cos x}{\sin x} \right]$$

Distributing the $(\sin x)^{g(x)}$ term:

$$f'(x) = g'(x)(\sin x)^{g(x)} \ln(\sin x) + g(x) \cos(x)(\sin x)^{g(x)-1}$$

Now we evaluate the limit as $x \rightarrow \pi$. First, find the limit of $g(x)$:

$$g(\pi) = (\cos \pi)^{\tan \pi} = (-1)^0 = 1$$

Let us analyze the two terms of $f'(x)$ as $x \rightarrow \pi$: For the first term, since $g(x) \rightarrow 1$, the expression involves $\sin(x) \ln(\sin x)$. It is a standard limit that $\lim_{u \rightarrow 0^+} u \ln(u) = 0$. Therefore, despite the presence of $g'(x)$ (which is bounded/finite), the first term evaluates to exactly 0. For the second term, we substitute the limits $g(\pi) = 1$, $\cos(\pi) = -1$, and the exponent $g(x) - 1 \rightarrow 0$:

$$\text{Second Term} = (1)(-1)(\sin \pi)^0 = (-1)(1) = -1$$

Combining both parts:

$$f'(\pi) = 0 - 1 = -1$$

The derivative at $x = \pi$ is exactly -1 .

3.2 Hard Derivative

Absolute Bounded Cubic Polynomial

Let $f(x) = ax^3 + bx^2 + cx + 5$. If $|f(x)| \leq |e^x - e^2|$ for all $x \geq 0$ and the maximum value of $|12a + 4b + c|$ is m , find m .

Solution: We are given the inequality $|f(x)| \leq |e^x - e^2|$ for all $x \geq 0$. Observe the behavior at $x = 2$. The right side of the inequality becomes $|e^2 - e^2| = 0$. Since $|f(2)| \leq 0$, we must strictly have $f(2) = 0$. Because $f(x)$ is continuously bounded by

a function that equals 0 at $x = 2$ and crosses the x-axis, $f(x)$ must be "sandwiched" at this point. Let us define the upper and lower bounding functions around $x = 2$: For $x > 2$, $e^x - e^2 > 0$, so:

$$-(e^x - e^2) \leq f(x) \leq e^x - e^2$$

Subtracting $f(2) = 0$ from all sides and dividing by $(x - 2) > 0$:

$$\frac{-(e^x - e^2)}{x - 2} \leq \frac{f(x) - f(2)}{x - 2} \leq \frac{e^x - e^2}{x - 2}$$

Taking the limit as $x \rightarrow 2^+$, the squeeze theorem yields:

$$-e^2 \leq f'(2) \leq e^2$$

Similarly, for $x < 2$, $e^x - e^2 < 0$, so $|e^x - e^2| = e^2 - e^x$. The inequality becomes:

$$-(e^2 - e^x) \leq f(x) \leq e^2 - e^x \implies e^x - e^2 \leq f(x) \leq e^2 - e^x$$

Dividing by $(x - 2) < 0$ reverses the inequality signs:

$$\frac{e^x - e^2}{x - 2} \geq \frac{f(x) - f(2)}{x - 2} \geq \frac{e^2 - e^x}{x - 2}$$

Taking the limit as $x \rightarrow 2^-$ yields $e^2 \geq f'(2) \geq -e^2$. Both one-sided limits confirm that $|f'(2)| \leq e^2$. Now, let's compute the generic derivative of the given polynomial $f(x)$:

$$f'(x) = 3ax^2 + 2bx + c$$

Evaluate this derivative at $x = 2$:

$$f'(2) = 3a(2)^2 + 2b(2) + c = 12a + 4b + c$$

The expression $|12a + 4b + c|$ is precisely $|f'(2)|$. We established that $|f'(2)| \leq e^2$. Thus, the absolute maximum value m that this expression can take is exactly e^2 .

$$m = e^2$$

Physics & Engineering

4.1 Mechanics

Semicircular Disk Rotation

A uniform semicircular disk has mass $M = 6 \text{ kg}$ and radius $R = 0.8 \text{ m}$. It rotates about a fixed frictionless axis perpendicular to its plane, through its geometric center. A constant force $F = 9 \text{ N}$ is applied at the rim, remaining tangential as the disk turns through an angular displacement of 5 rad . Starting from rest, find its final angular speed ω in rad s^{-1} .

Solution: The moment of inertia I of a uniform semicircular disk rotating about an axis perpendicular to its plane and passing through its geometric center (the center of the straight edge) is identical to the formula for a full disk of the same mass M and radius R . Thus:

$$I = \frac{1}{2}MR^2$$

Substitute the given values:

$$I = \frac{1}{2}(6 \text{ kg})(0.8 \text{ m})^2 = 3 \times 0.64 = 1.92 \text{ kg m}^2$$

The constant tangential force F applied at the rim creates a constant torque τ :

$$\tau = F \times R = 9 \text{ N} \times 0.8 \text{ m} = 7.2 \text{ N m}$$

The total work W done by this torque over the angular displacement $\Delta\theta = 5 \text{ rad}$ is:

$$W = \tau\Delta\theta = 7.2 \times 5 = 36 \text{ J}$$

According to the Work-Energy Theorem for rotational motion, the work done equals the change in rotational kinetic energy. Since the disk starts from rest ($\omega_i = 0$):

$$W = \frac{1}{2}I\omega^2$$

$$36 = \frac{1}{2}(1.92)\omega^2 = 0.96\omega^2$$

Solving for the square of the angular speed:

$$\omega^2 = \frac{36}{0.96} = \frac{3600}{96} = 37.5$$

Taking the square root gives the final angular speed:

$$\omega = \sqrt{37.5} = \sqrt{\frac{150}{4}} = \frac{5\sqrt{6}}{2} \approx 6.12 \text{ rad s}^{-1}$$

4.2 Quantum Mechanics

Hydrogen-Like Ion Orbital Speed

A hydrogen-like boron ion B^{4+} has $Z = 5$. In the Bohr model, its orbital radii obey $r_n = \frac{n^2 a_0}{Z}$. Using Coulomb force as the centripetal force, find the orbital speed ratio $\frac{v_2}{v_3}$.

Solution: According to classical mechanics applied to the Bohr model, the Coulomb force provides the centripetal force required to keep the electron in a circular orbit of radius r_n and speed v_n :

$$\frac{kZe^2}{r_n^2} = \frac{m_e v_n^2}{r_n}$$

Simplifying this equation for v_n^2 , we get:

$$v_n^2 = \frac{kZe^2}{m_e r_n}$$

We are given that the orbital radius scales proportionally to the square of the principal quantum number n :

$$r_n = \frac{n^2 a_0}{Z} \implies r_n \propto n^2$$

Substituting this proportionality into the velocity squared equation yields:

$$v_n^2 \propto \frac{1}{n^2}$$

Taking the square root demonstrates that the orbital speed v_n is inversely proportional to n :

$$v_n \propto \frac{1}{n}$$

Notice that this ratio is entirely independent of the atomic number Z . We evaluate the ratio of the speeds at $n = 2$ and $n = 3$:

$$\frac{v_2}{v_3} = \frac{1/2}{1/3} = \frac{3}{2} = 1.5$$

4.3 Electric Fields

DC Parallel Circuit Power

An ideal DC source is connected across two parallel branches. The first branch contains 8Ω and 12Ω resistors in series. The second branch contains a 5Ω resistor in series with another resistor. The voltage across the 8Ω resistor is 9 V , and the 5Ω resistor dissipates 20 W . Find the total power supplied by the source in watts. Give your answer to two decimal places.

Solution: Let V_s be the voltage of the ideal DC source. **Branch 1 Analysis:** The total resistance of the first branch is the sum of the series resistors:

$$R_1 = 8\Omega + 12\Omega = 20\Omega$$

We are given the voltage across the $8\ \Omega$ resistor is 9 V . Using Ohm's Law, the current I_1 flowing through Branch 1 is:

$$I_1 = \frac{V_8}{8} = \frac{9}{8} = 1.125\text{ A}$$

Since the entire branch is connected directly across the source, the source voltage V_s is:

$$V_s = I_1 \times R_1 = 1.125\text{ A} \times 20\ \Omega = 22.5\text{ V}$$

Branch 2 Analysis: The $5\ \Omega$ resistor in the second branch dissipates 20 W of power. We can use the power formula $P = I^2 R$ to find the current I_2 in Branch 2:

$$20 = I_2^2 \times 5 \implies I_2^2 = 4 \implies I_2 = 2\text{ A}$$

Total Power Analysis: The total current I_{total} supplied by the DC source is the sum of the currents in both parallel branches:

$$I_{\text{total}} = I_1 + I_2 = 1.125\text{ A} + 2\text{ A} = 3.125\text{ A}$$

The total power P_{total} supplied by the source is the product of the source voltage and the total current:

$$P_{\text{total}} = V_s \times I_{\text{total}} = 22.5\text{ V} \times 3.125\text{ A}$$

$$P_{\text{total}} = 70.3125\text{ W}$$

Rounding to two decimal places as requested, the total power is 70.31 W .